Pathophysiology of Chronic Venous Insufficiency and Venous Leg Ulcers

All structures of the venous system may contribute to chronic venous insufficiency (CVI) and consequently to the development of venous leg ulcers.

Varices initially develop due to alterations in the vein wall, leading to valvular incompetence in both the descending and ascending venous pathways.

While some studies report a higher prevalence of varicose veins in women, this may reflect selection bias, as women are more likely to seek medical evaluation for cosmetic concerns. Population-based studies generally show no significant sex difference in varicose veins. However, advanced chronic venous disease (CVD) affects both sexes equally and can progress to leg ulceration—the end stage of CVI. Venous leg ulceration has a lifetime prevalence of 1% and an active prevalence of 0.3% in the adult Western population, with increasing incidence with age. Ulceration in individuals under 60 is uncommon and often associated with severe deep venous insufficiency.

As CVI progresses, venous hypertension extends to the venules and capillaries. Elevated intracapillary pressure increases capillary filtration, resulting in fluid leakage and edema. Concurrently, high molecular weight proteins such as fibrin escape into the pericapillary space, forming visible "fibrin cuffs" around capillaries.

Initially, it was hypothesized that these fibrin cuffs impair oxygen diffusion, causing tissue anoxia and ulceration. However, this theory was challenged by findings of fibrin cuffs in other skin conditions without reduced transcutaneous oxygen tension. Alternative mechanisms include leukocyte trapping in capillaries and the release of free radicals, contributing to tissue damage.

The transmission of high venous pressure to the dermal microcirculation triggers a chronic inflammatory response. This involves cytokine and growth factor release, promoting leukocyte migration into the interstitium and perpetuating inflammation. This process underlies the dermal fibrosis and tissue remodeling characteristic of CVI.

Microscopic studies have also revealed capillary thrombosis in white atrophy and other cutaneous manifestations of CVI, suggesting microvascular occlusion as another potential contributor to ulcer formation. More recently, it has been proposed that fibrin cuffs may sequester growth factors rather than oxygen, thereby impairing wound healing by limiting growth factor availability at the ulcer site.

These interrelated mechanisms are integrated into the **Rotterdam model**, which summarizes the multifactorial pathogenesis of venous leg ulcers.

Some authors suggest that **biofilm formation** may also play a significant role in delaying healing in chronic venous ulcers.

Risk Factors

Obesity is a recognized risk factor for venous leg ulcers. It can induce venous hypertension and subsequent ulceration even in the absence of valvular incompetence or venous obstruction. **Dependency**, due to immobility, impairs muscle pump function and leads to venous stasis and hypertension. In both scenarios, venous valves may remain structurally patent, emphasizing the need for comprehensive diagnostic evaluation.

Therefore, clinical examination should be supplemented with technical investigations, particularly **duplex ultrasound**, ideally performed by a physician in collaboration with a technician, to accurately assess venous function and guide management.

Symptomatology

Venous leg ulcers may arise spontaneously or follow minor trauma. Patient symptoms can range from mild to severe. Contrary to traditional teaching, venous leg ulcers **can be painful**, particularly during the ulcerative phase of white atrophy or when complicated by infection.